

AI Study Material

RAG, Embeddings, and Vector Databases

Use this PDF as study material for your AI Study Agent Phase 8 test. The three topics are separated so the test generator has distinct concepts to work from.

Topic 1 — RAG (Retrieval-Augmented Generation)

- RAG stands for Retrieval-Augmented Generation.
- A RAG system retrieves relevant information from a knowledge source and gives that information to an AI model as context.
- The retrieved information helps the model answer questions using the available study material.
- A simple RAG flow is: user question → retrieval → relevant context → AI model → answer.
- RAG is useful when an application needs answers grounded in documents or other provided knowledge.

Topic 2 — Embeddings

- An embedding is a numerical representation of text or another piece of information.
- Embeddings allow a system to compare semantic meaning.
- Texts with similar meanings can have embeddings that are close to each other in vector space.
- Embeddings are commonly used in semantic search and retrieval systems.
- Two sentences can use different words but still have similar embeddings when they express a similar idea.

Topic 3 — Vector Databases

- A vector database is designed to store and search vector representations such as embeddings.
- A vector database can compare a query vector with stored vectors and retrieve similar items.
- Vector databases are useful for semantic search because they can search based on meaning rather than exact keyword matches.
- In a RAG system, document chunks can be converted into embeddings and stored in a vector database.
- A user question can be embedded and used to find relevant document chunks.

How They Work Together

RAG is the overall approach for grounding an AI answer in retrieved information. Embeddings are numerical representations used to compare semantic meaning. A vector database stores and searches those vector representations.

Topic	Main idea
RAG	Retrieve useful information and give it to the AI model as context.
Embeddings	Represent information numerically for semantic comparison.
Vector Databases	Store and search vectors for similarity.

Phase 8 Test Target

Generate 5 medium questions from this material. Ideally, questions should cover multiple topics, for example:

- Question 1 → RAG
- Question 2 → Embeddings
- Question 3 → Vector Databases
- Question 4 → RAG
- Question 5 → Embeddings